

Supplementary Material for “Speculative GHR Forwarding: Eliminating Stale Branch-Predictor State in Deep FPGA Pipelines”

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This supplement collects corroborating figures and tables referenced from the main paper, plus a Mechanism-B susceptibility screen on two additional workloads. Most are reproduced from the same measured runs; the cache-miss table is modeled, as noted there. Nothing here is required to follow the main argument, which is why these floats are placed out of line. Section numbers (S...) and float labels (Fig. S..., Table S...) match the citations in the main text.

1 Area Scaling with Depth

Figure S1 shows the slice LUT and flip-flop growth across the five pipeline depths. The 4-to-8-stage increase is 622 LUTs (8.6%) and 638 FFs (7.9%); DSP usage is constant at 12 blocks. The growth is gradual and sub-linear in the added pipeline registers, because the dominant cost is the shared functional-unit logic (ALU, MDU, register file), identical across depths; only the inter-stage registers and the slightly larger forwarding/hazard logic grow.

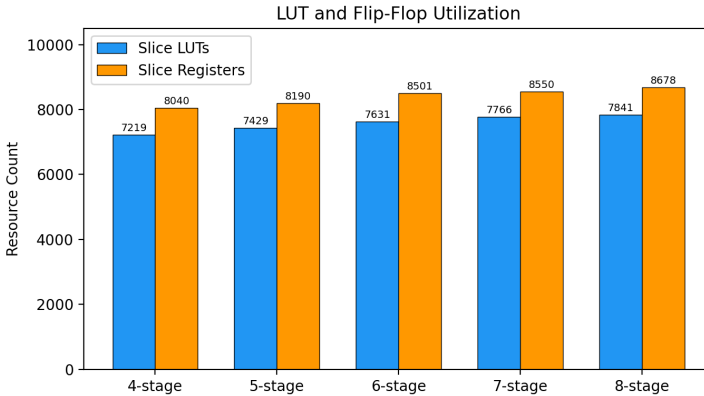


Fig. S1. Slice LUT and flip-flop utilization across pipeline depths on Artix-7 XC7A35T. The 4-to-8-stage increase is 622 LUTs (8.6%) and 638 FFs (7.9%); DSP usage is constant at 12 blocks.

2 CPI Model Residuals

Figure S2 plots the CPI model residuals (measured minus predicted) across depths. They have mean 0 and standard deviation 0.038 CPI with no depth-dependent pattern; the largest residual (+0.086) is on 6-stage CoreMark, reflecting the smooth $M_w(D)$ slightly under-capturing the depth-6 misprediction jump.

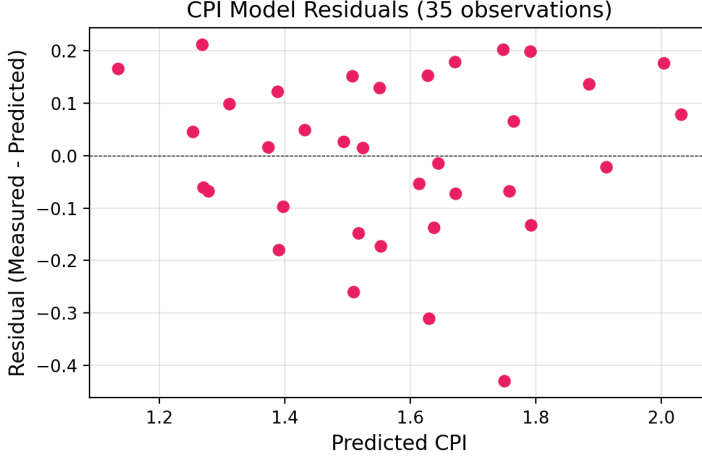


Fig. S2. CPI model residuals (measured minus predicted) across depths. Mean 0, standard deviation 0.038 CPI, with no depth-dependent pattern; the largest residual (+0.086) is on 6-stage CoreMark.

3 BRAM Instruction Memory Effect

Figure S3 compares $F_{\max}$ with LUT-RAM versus BRAM instruction memory across the five depths. With BRAM the deep tier rises to 121.5 MHz (6-stage), 120.9 MHz (7-stage), and 121.2 MHz (8-stage); the IF1/IF2 split absorbs the BRAM's 1-cycle read latency, so the 7/8-stage rise to near the 6-stage, while the shallow tier holds (4-stage 69.3, 5-stage 73.7 MHz). Mapping to BRAM shifts frequency but not the CPI ordering, so the 6-stage stays the throughput optimum.

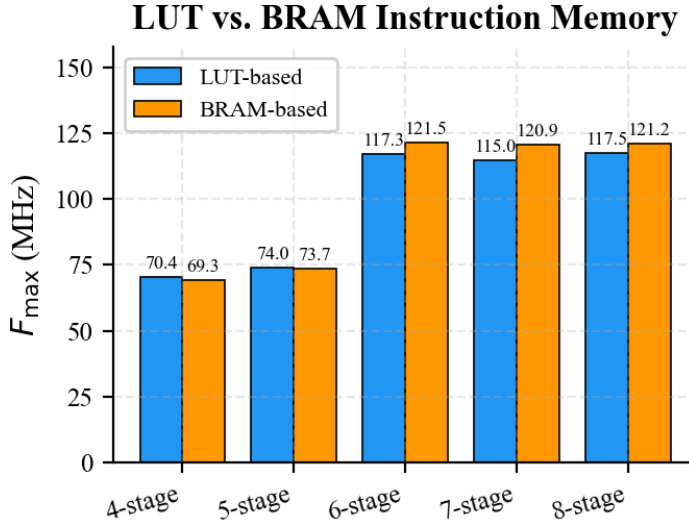


Fig. S3. $F_{\max}$ with LUT-RAM versus BRAM instruction memory across the five depths. The IF1/IF2 split absorbs the BRAM read latency, so the 7/8-stage rise to near the 6-stage; the 4/5-stage tier is unchanged.

4 Power and Efficiency

Figure S4 shows the on-chip power breakdown across depths under workload-specific (CoreMark SAIF) switching activity, and Figure S5 the resulting efficiency. Total power falls with depth from 0.235 W (4-stage) to 0.217 W (8-stage) because lower IPC reduces average toggling. The 6-stage is the most energy-efficient depth at 286 MIPS/W, as well as the highest-throughput. These are Vivado vector-based estimates at “Medium” confidence, not physical measurements; we rely on the relative ordering across depths, not the absolute watts.

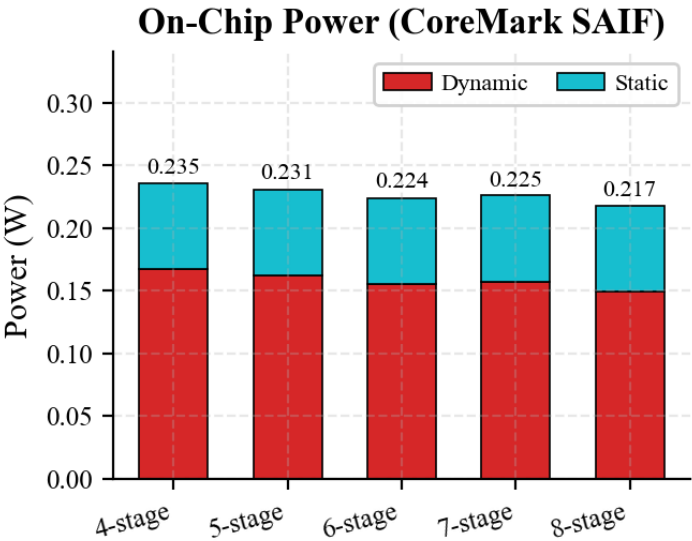


Fig. S4. On-chip power (dynamic and static) across the five depths under workload-specific (CoreMark SAIF) switching activity. Total power falls with depth as lower IPC reduces average toggling.

5 Cache-Miss Sensitivity (Modeled)

Table S1 works the 7-stage CoreMark case through four modeled cache-miss scenarios. SGF’s absolute cycle saving (159,932 cycles) is miss-rate-independent, so its relative benefit shrinks as memory stalls grow to dominate CPI. These are analytical projections, not measured against a real hierarchy.

Table S1. SGF throughput improvement under modeled cache-miss rates (7-stage CoreMark). The absolute cycle saving from SGF (159,932 cycles) is constant, so the relative improvement decreases as memory stalls grow to dominate CPI.

Miss scenario	Miss CPI	Total CPI (base)	Total CPI (SGF)	Δ CPI	Rel. improvement
No misses (this study)	0	2.02	1.96	−0.06	3.0%
1% miss, 20-cycle penalty	0.20	2.22	2.16	−0.06	2.7%
2% miss, 50-cycle penalty	1.00	3.02	2.96	−0.06	2.0%
5% miss, 100-cycle penalty	5.00	7.02	6.96	−0.06	0.9%

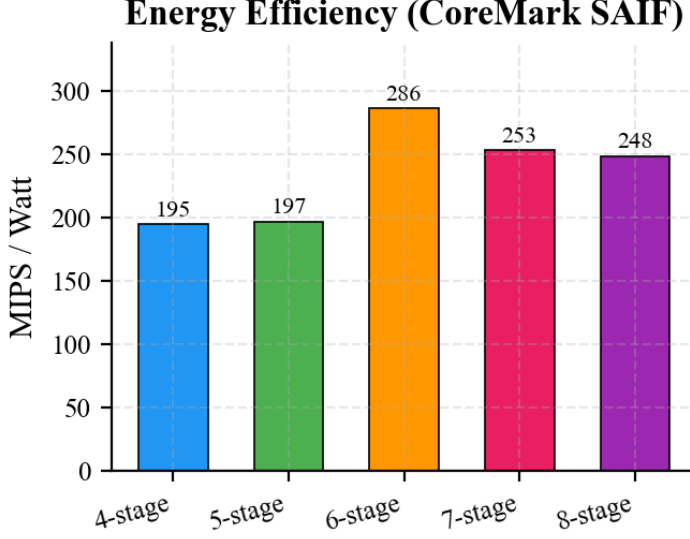


Fig. S5. Power efficiency (MIPS/W) across the five depths, from SAIF-based power and CoreMark throughput. The 6-stage is the most efficient at 286 MIPS/W.

6 Published-Cores Context

Figure S6 places our five variants against published Artix-7 RISC-V soft cores. It is included only to confirm our research vehicle sits in a realistic design-space region (70–118 MHz); the cores differ in ISA subset, execution model, predictor, and memory, so their frequency gaps reflect those confounders, not pipeline depth. We make no cross-project performance claim.

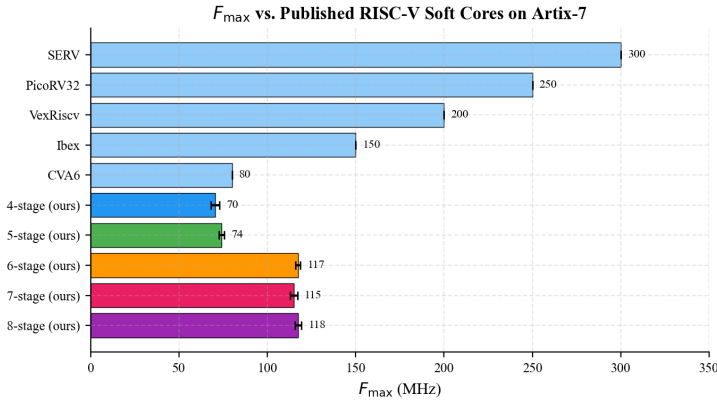


Fig. S6. $F_{\max}$ of our five variants against published Artix-7 RISC-V soft cores. Included only to confirm our research vehicle sits in a realistic design-space region; the cores differ in too many variables to support a cross-project performance comparison.

7 CoreMark Detail Across Depths

Table S2 gives the full official CoreMark figures across the five depths. The CPI and misprediction columns also appear in the main paper; the cycle and CoreMark/MHz rows are reported here in full.

Table S2. Official EEMBC CoreMark results across the five pipeline depths (10 iterations, PERFORMANCE_RUN). All variants execute identical instruction counts (2,893,145) and branch counts (719,810). Cycles and mispredictions are from RTL simulation; CoreMark/MHz uses the post-place-and-route $F_{\max}$.

Metric	4-stg	5-stg	6-stg	7-stg	8-stg
Cycles	4,481,534	4,724,937	5,310,372	5,856,929	6,323,069
Mispredictions	145,735	145,735	154,077	147,760	147,760
Mispredict rate	20.2%	20.2%	21.4%	20.5%	20.5%
CPI	1.54	1.63	1.83	2.02	2.18
CoreMark/MHz	2.23	2.12	1.88	1.71	1.58

8 CoreMark CPI Ablation

Table S3 decomposes CoreMark CPI into the four additive terms of the main paper’s CPI model equation at each depth. The summed prediction matches measured CPI within the model residual (≤ 0.09).

Table S3. CPI ablation for CoreMark: decomposition into the architectural cost components of the CPI model equation (main paper) at each pipeline depth. Each row is one term. The sum matches measured CPI within the model residual (≤ 0.09).

CPI component	4-stg	5-stg	6-stg	7-stg	8-stg
Ideal throughput	1.00	1.00	1.00	1.00	1.00
Branch flush ($M \cdot P/N$)	0.10	0.10	0.16	0.20	0.20
Load-use stall ($\alpha \cdot L$)	0.00	0.15	0.15	0.15	0.22
IF2 bubble ($\beta \cdot B \cdot f_b$)	0.00	0.00	0.00	0.28	0.28
Workload base (γ_w)		0.45 (depth-invariant)			
Model prediction	1.55	1.70	1.76	2.08	2.15
Measured CPI	1.54	1.63	1.83	2.02	2.18

9 Mechanism-B Susceptibility by Inter-Branch Distance

Table S4 classifies the workloads by inter-branch distance (IBD) and observed Mechanism B behavior. Four workloads exhibit Mechanism B. CoreMark and statemate are from the main suite; huffbench and sglib-combined are from the susceptibility screen (Section 11). All four have IBD below 10 and data-dependent outcomes. Workloads with larger IBD or highly biased outcomes show no depth-dependent variation.

10 Pipeline Structure and Hazard Costs

Table S5 lists the per-event hazard costs across the five depths (architectural parameters, not measured data). The bottom row, predictor update latency, is the central variable for Mechanism B.

Table S4. Mechanism B susceptibility classified by inter-branch distance (IBD) and branch outcome predictability. huffbench and sglib-combined are from the susceptibility screen (Section 11); their inflation is measured against the 4-stage reference.

Workload	IBD	Outcome type	Mech. B?
CoreMark	4.0	Data-dependent	Yes (+5.7%)
statemate	8.4	Data-dependent	Yes (+10%)
huffbench	3.3	Data-dependent	Yes (+2.0%)
sglib-combined	4.0	Data-dependent	Yes (+2.4%)
edn	8.8	Mixed	No
aha-mont64	8.7	Biased loops	No
Diagnostic	3.8	Structured	No
Dhrystone	7.3	Structured	No
crc32	23.0	Highly biased	No

Table S5. Pipeline structure and per-event hazard costs across depths (architectural parameters, not measured data). Predictor update latency (bottom row) is the central variable for Mechanism B.

Property	4-stg	5-stg	6-stg	7-stg	8-stg
Pipeline registers	3	4	5	6	7
Forwarding sources	1	2	3	3	3
Load-use stall	0	1 cy	1 cy	1 cy	1–2 cy
Branch penalty	2 cy	2 cy	3 cy	4 cy	4 cy
IF2 prediction bubble	0	0	0	1 cy	1 cy
Predictor update latency	2	2	3	4	4

11 Mechanism-B Susceptibility Screen

The main suite has two Mechanism-B workloads. To broaden the evidence, we screened two further low-IBD, data-dependent Embench-IoT kernels, huffbench (IBD 3.28) and sglib-combined (IBD 4.00). Table S6 reports their misprediction counts across depth and under SGF. Both rise over the 4-stage reference into the deep tier, the Mechanism-B signature, and SGF reduces the count at the 6-, 7-, and 8-stage.

These kernels need a 16 KB data memory to fit, whereas the main suite uses 8 KB. Data-memory size changes synthesis area, frequency, and power, but not the control-path misprediction counts measured here. We also use the 4-stage as the shallow reference, not the 4/5-stage pair used for the main suite. These kernels have branches that depend on nearby loads. The 5-stage load-use stall shifts those branches by one cycle and lowers its count slightly, while the 4-stage has no such stall. The 4-stage is therefore the cleaner baseline. This effect is confined to the shallow tier and does not change the deep-tier rise or the SGF reduction.

Under SGF, sglib-combined’s 8-stage reduction (17.5%) is smaller than its 7-stage (21.8%), a wider 7-to-8 gap than the main suite shows. The same load-dependent branch timing acts through the 8-stage’s dual load-use stall.

The bimodal and PHT-capacity controls that isolate Mechanism B causally were run on the main suite, not on these kernels. The predictor-family (tournament, TAGE) and -O2 robustness studies likewise use the main suite. The screen evaluates the default gshare predictor at -O1, and tests only whether more workloads show the effect. Here the depth-dependent rise and the SGF response are consistent with that mechanism, on the same gshare predictor. As in the main suite, the SGF

reduction exceeds the baseline inflation because SGF removes the full update latency, not only its depth-dependent slice.

Table S6. Mechanism-B susceptibility screen: misprediction counts (RTL simulation) for two additional Embench-IoT kernels across depth and under SGF. Instruction and branch counts are identical across depths (huffbench 2,098,752/638,859; sglib-combined 2,431,970/607,229). The 7- and 8-stage baselines are equal, as in the main suite.

Workload	IBD	Baseline mispred			SGF mispred		
		4-stg	6-stg	7/8-stg	6-stg	7-stg	8-stg
huffbench	3.28	187,420	191,088	191,090	109,903	109,903	110,192
sglib-combined	4.00	171,291	175,477	175,043	140,026	136,959	144,334

SGF reduction vs. the same-depth baseline. huffbench: -42.5% , -42.5% , -42.3% at 6/7/8-stage. sglib-combined: -20.2% , -21.8% , -17.5% at 6/7/8-stage. CPI falls at every depth (huffbench $1.86 \rightarrow 1.74$ at 6-stage, $2.12 \rightarrow 1.96$ at 7-stage).